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https://github.com/vpik009/emp/tree/main

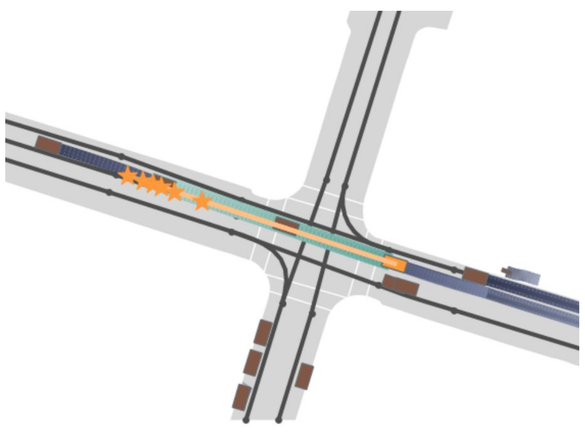
RL#5 Small Footprint Motion Prediction with Teacher-Student Learning

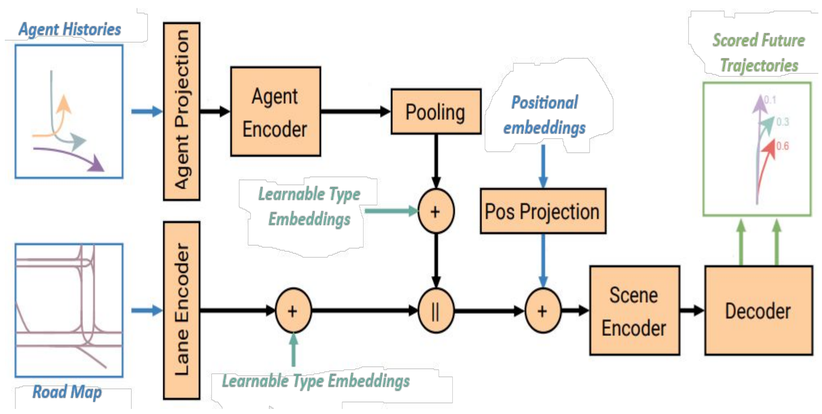
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Introduction

The authors of [1] present **EMP**: a model for *motion prediction* with strong performance on the ArgoVerse2 dataset [2].





the architecture of EMP and all our student models ([1, Figure 2])

Since EMP has 3.2 million parameters, we investigate Student-Teacher methods to obtain smaller student models that still perform well in key metrics.

We consider four student models that vary along two dimensions:

- Size: *small* (700k parameters) and *tiny* (300k parameters)
- Shape: *thin* and *wide* models, which differ in their embedding dimension and transformer depth

Relational Knowledge Distillation

Improves student learning by transferring teacher relational knowledge [3]: matches pairwise distances and triplet angles via Huber loss across batches, to explicitly preserve embedding geometry:

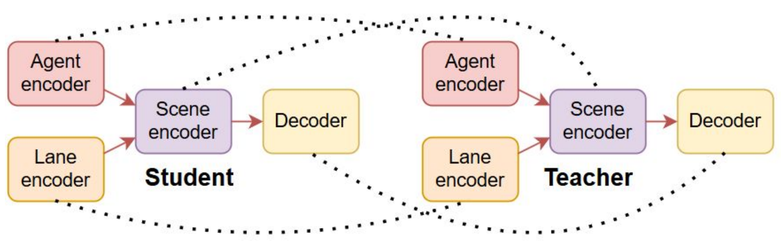
$$L = L_{task} + \lambda_d L_{dist} + \lambda_a L_{ang}$$

$$L_{dist} = \frac{1}{|\mathcal{B}|} \sum_{(i,j)\in\mathcal{B}} \text{Huber}\Big(d(x_i^s, x_j^s) \; - \; d(x_i^t, x_j^t) \Big)$$

$$L_{\text{ang}} \; = \; \frac{1}{|\mathcal{B}|} \sum_{(i,j,k)\in\mathcal{B}} \text{Huber}\Big(\cos\theta_s^{(i,j,k)} \; - \; \cos\theta_t^{(i,j,k)} \Big)$$

Hint-based Distillation

Based on *FitNets* [4]. In several stages, the outputs of teacher and student are trained to match each other via pooling across the embedding dimension. Final loss depends on both true targets and teacher output.



Trajectory Distance Distillation

Use the training data to train the student and the teacher model to guide the student towards teacher predictions after 20 epochs.

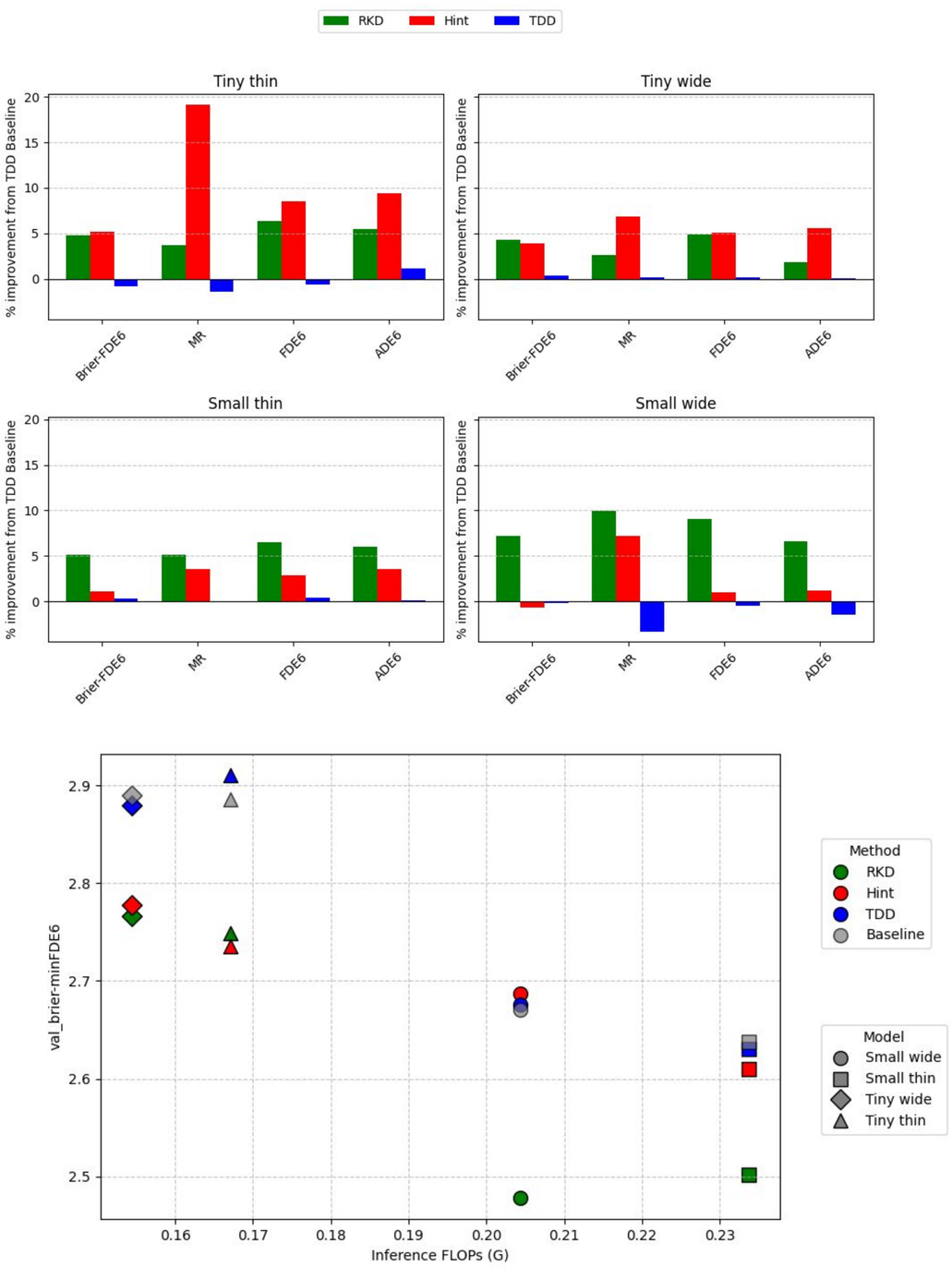
$$distance(t_{traj}, s_{traj}) = \sqrt{\sum_{t=1}^{T} [(x_t^{teacher} - x_t^{student})^2 + (y_t^{teacher} - y_t^{student})^2]}$$

Then weight that using product of the student and teacher confidence on that trajectory and append to standard loss:

$$L = (1 - \alpha) \cdot L_{stand} + \alpha \cdot L_{distill}$$

An adaptation of the method discussed in [5]

Quantitative Results



Discussion

Qualitative results:

- Baselines already quite strong; variations across methods, metrics, and student model types
- RKD: Consistent improvements, although better results for larger student models
- Hint: Better performance with tiny than small, thin than wide models (the hints help training deep models)
- TDD: Gains vary with model size and capacity. No consistent gains across a specific metric across the models

Future work:

- RKD: Modeling the teacher's embedding geometry for non-focal agents and scene objects
- Hint: Try to find working hint methods beyond pooling across the embedding dimension
- TDD: Adjust alpha dynamically based on the validation performance
- TDD: Experiment with confidence weighting methods that prioritize those trajectories that the teacher finds important

References

[1] Zhang, L., Wang, Y., & Chen, H. (2023). *Efficient Motion Prediction: A Lightweight & Accurate Trajectory Prediction Model With Fast Training and Inference Speed*. arXiv preprint arXiv:2306.00987.
[2] B. Wilson et al. (2021). "Argoverse 2: Next Generation Datasets for Self-driving Perception and Forecasting", *Proc. of the Neural Information Processing Systems Track on Datasets and Benchmarks* (NeurIPS Datasets and Benchmarks 2021).
[3] Park, W., Kim, D., Lu, Y., & Cho, M. (2019). *Relational Knowledge Distillation*. arXiv preprint arXiv:1904.05068.
[4] Romero, A., Ballas, N., Kahou, S. E., Chassang, A., Gatta, C., & Bengio, Y. (2015). *FitNets: Hints for Thin Deep Nets*. arXiv preprint arXiv:1412.6550.
[5] Hinton, G., Vinyals, O., & Dean, J. (2015). *Distilling the Knowledge in a Neural Network*. arXiv:1503.02531.